DS → SG 2/25/86 CC: DRS. JACOBSON, SOMMERS
LJ

ST. MARY'S HOSPITAL MEDICAL SCHOOL

Norfolk Place London W2 1PG
Telephone: 01-723 1252

PROFESSOR OF BIOCHEMICAL PHARMACOLOGY: Robert L. Smith DSc, PhD.

JRI/HEJ

18 February 1986

Dr D Stone
Council for Tobacco Research - U.S.A. Inc.
900 Third Avenue
New York
NY 10022
U.S.A.

Dear Dr Stone

Re: Grant Application # 1955 - Edward Calabrese, Ph.D.

Thank you for sending me the above application and enclosures. I apologise for the short delay, but it has taken me some time to go through it thoroughly. I have organised my comments into the areas which I think are pertinent to any grant application. I hope that these will help you reach a decision.

(i) Aims of the project

The applicant seeks to confirm and extend the findings published by us in 1984 that there is a possible inherited and metabolic susceptibility to lung cancer in smokers. They do not propose to deviate significantly from our own study design with the exception of selection of different types of controls and a broader data analysis.

(ii) Work which led up to the project

A decade of research effort almost exclusively in our own laboratories has led to a greater understanding of the genetics of human drug oxidation processes. At least three independent allelomorphic genes have been described, those regulating debrisoquine 4-hydroxylation and carbocisteine S-oxidation - the applicant refers only to the former of these. Over the years, it has been shown that many oxidative pathways are influenced to a greater or lesser degree by the gene which regulates debrisoquine 4-hydroxylation. In addition, there are many pharmacokinetic, therapeutic and toxicological sequelae of the polymorphism. Our own biochemical epidemiological studies in lung cancer patients and controls form the basis of Calabrese's application. The lack of perception of the relevant literature to date shown by the applicant in my view detracted considerably from his case. There are several misciations of the literature and in addition Dr Calabrese has found it necessary to quote directly from our papers rather than paraphrasing them. Most importantly, none of the work which has led up to the proposed project was carried out by Dr Calabrese and his collaborators.

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